

Black Bean Prep

RS3 + anthocyanin matrix → soak → boil → chill → freeze

① Overnight Soak

	3-Month Batch	Weekly Mode
Soak liquid	Baking soda water ¼ tsp per quart	Filtered water + 1 tsp sea salt
Time	Overnight (8–12h) at room temp	12–16h at room temp
After	Drain + rinse thoroughly — discard soak water	Drain + rinse thoroughly
Why baking soda	Reduces oligosaccharides ~70% + softens skins. NEVER cook in soak water — degrades aquafaba emulsifiers.	Salt alone softens skins less aggressively. Same discard rule applies.

Optional kombu: add a 2–3 inch strip of dried kombu to the SOAK water (remove before cooking). Kombu contains alpha-galactosidase — same mechanism as hing — breaks down raffinose and stachyose oligosaccharides that cause gas. Complements rather than duplicates the hing in the boil stack. Not for daily use if thyroid-sensitive (iodine load).

② Ayurvedic Boil Stack

Add ALL aromatics to COLD water before turning on heat. Cook in a separate pot from chickpeas — keep liquid completely separate.

Ingredient	Amount (weekly 2–3 cups dry)	Why
Hing (asafoetida)	■ tsp	Anti-flatulent (gentler than chickpeas — do NOT use ¼ tsp)
Fresh ginger (smashed)	1 inch knob	Pro-kinetic, anti-nausea — whole for easy removal



Whole cumin seeds (jeera)	1 tsp	Digestive enzyme stimulation
Ajwain	½ tsp	Thymol — strongest carminative in the stack
Fennel seeds	1 tsp	Anethole — carminative, anti-spasmodic
Coriander seeds	1 tbsp	Digestive, pairs with cumin
Bay leaves	2	Linalool + eugenol — anti-inflammatory
Whole cloves	3–4	Eugenol — antimicrobial, anti-inflammatory
Green cardamom (crushed)	1 tsp / ~6 pods	Cineole — digestive, carminative
Black cardamom (crushed)	½ tsp / ~1 pod	Smoky digestive depth
Star anise	½ tsp / ~1 point	Anethole — carminative, stronger than fennel
Ceylon cinnamon stick	1 small	Cinnamaldehyde — glycemic anchor
Onion	1 medium, quartered OR 2 tbsp onion powder	Quercetin + prebiotic FOS. Whole preferred — powder clouds the liquid
Garlic (lightly crushed)	6 whole cloves	Aromatic infusion (ajoene/DAS) — reserve for mash after boil
Black pepper (coarse crack)	1 tbsp	Piperine into liquid — fat-soluble absorption enhancer. Coarse prevents gritty texture.
Sea salt	1 tsp per quart — LAST 15 MIN ONLY	Early salt = tough skins. Last 15 min opens pectin so aromatics penetrate.

Boil: bring to boil → skim foam → simmer 60–90 min until tender but not mushy.



Black Bean Prep (continued)

Anthocyanin liquid · chill → freeze · day of fat bloom

③ Anthocyanin Liquid — Reserve Before Draining

Step	Action
Reserve	Before draining — ladle out the dark purple cooking liquid into a separate bowl
Do NOT	Do NOT reduce or boil down — anthocyanins (cyanidin-3-glucoside) concentrate quickly and turn bitter
Cool	Cool at room temp as-is after straining beans out
Freeze	2 tbsp cubes (regular tray) + some ¼ cup portions. Label: 'Black Bean Liquid'
Use	Recovery bowl (2 tbsp into yogurt at 17:50) · Refried beans base · Samosa filling mash (2 tbsp) The dark purple color IS the signal compound — not a flaw.

④ Chill → Freeze (RS3 Formation)

Mode	After draining	Chill	Freeze
Weekly	Spread on stainless tray · cool · refrigerate	24h RS3 build	After allicin pulse + 2nd chill
3-Month batch	Spread on sheet pans · cool to 130°F · fold in ACV + pinch salt → portion into 1.5-cup bags immediately	Single 12–24h chill in portioned bags	Transfer to chest freezer next morning

Single chill builds same RS3 as double-chill (starch retrogradation is indifferent to what else is in the container). Double-chill for weekly mode already has that structure; batch mode eliminates the re-warm loop across 5 kg.

⑤ Day-Of: Allicin Pulse + Fat Bloom + Finishers

Step	Action	Temp / Rule
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Warm	Gently warm beans in skillet on LOW	Stay below 130°F — RS3 collapses above this
Allicin pulse	Drop 1 rested garlic cube (or crush + rest 10 min fresh) Fold through warm beans with ACV + pinch salt	Fold gently — don't mash
Fat bloom	Drizzle 1–2 tsp EVOO or avocado oil over warm beans Add day-of spices: Kashmiri chili powder + cumin + freshly cracked black pepper Stir to coat — fat carries fat-soluble compounds (capsaicin, piperine, curcumin)	Add AFTER allicin cube EVOO oxidizes frozen — always day-of only
Finishers (last second)	Black salt (kala namak) · fresh lemon squeeze · fresh cilantro (chopped)	NEVER pre-assemble or freeze these in Volatile oils + sulfurous compounds degrade

■ **130°F CEILING** — never microwave on high or re-boil after RS3 is built. Reheat gently on low (warm to touch). Salt **LAST 15 MIN** during boil — early salt toughens skins and slows cooking by 20–30%.

